

**Ain Shams University
Faculty of Engineering
International Credit Hours Program**

CSE241 – Object-Oriented Programming

**Project Title: Hotel Management System
Milestone 2**

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1. Introduction

Milestone 2 builds upon the solid OOP backend established in Milestone 1 by adding a complete JavaFX-based Graphical User Interface (GUI). The goal of this milestone was to create an intuitive, visually consistent desktop application that allows guests, receptionists, and administrators to interact with the Hotel Management System through a fully functional set of screens.

All user interface screens were designed and implemented using JavaFX with FXML layout files, and a shared CSS stylesheet was applied throughout to ensure a professional and consistent appearance. The system supports two distinct user roles – Guest and Staff (Admin/Receptionist) – each with their own login flow and dashboard.

2. System Overview

The Hotel Management System GUI was implemented entirely in JavaFX with FXML-based layouts. The application consists of the following major screen groups:

- Guest Side: Welcome Screen, Guest Login, Guest Registration, Guest Dashboard, Make Reservation, Available Rooms, Manage Reservations, Cancel Reservation, and Checkout.
- Staff Side: Staff Login, Receptionist Dashboard (view/confirm/cancel/check-in/checkout reservations, view guests and rooms), and Admin Dashboard (add room, add room type, add amenity, add receptionist).

All screens share a consistent visual design using a warm color palette (primary color: #E06D67, background: #FFFCF5) defined in a single CSS stylesheet. Navigation between screens is handled through scene switching.

3. JavaFX Graphical User Interface

3.1 Technology Requirements

The GUI was built exclusively using JavaFX as required. All layouts, screens, and event handling use the JavaFX framework. The use of Swing was deliberately avoided throughout the project.

- GUI Framework: JavaFX
- Layout Files: FXML (all screens use FXML)
- Styling: CSS stylesheet (styles.css) applied globally
- IDE: IntelliJ IDEA

3.2 CSS Stylesheet

A single CSS stylesheet (styles.css) is applied across all screens to maintain a consistent and professional appearance. The stylesheet defines the following key style classes:

- .root – Sets the global background color to #FFFCF5 (warm off-white).
- .title-label – Bold 32px dark text for page headings.
- .field-label – Small 12px labels for input fields.
- .input-field – Rounded white input boxes with border color #D0D0D0, changing to #E06D67 on focus.

- `.btn-main` – Primary action buttons with #E06D67 background, white text, rounded corners. Darkens on hover to #C95A54.
- `.btn-secondary` – Outlined buttons with transparent background and red border/text that fills on hover.
- `.btn-secondary2` – Outlined secondary style with white text for use on colored backgrounds.
- `.status-label` – Red error/status messages for validation feedback.

3.3 Screen Descriptions and Sample Output

3.3.1 Welcome Screen

The application opens with a visually striking Welcome Screen that displays a full-background hotel photo of HMS Hotel, Cairo, Egypt. A centered frosted card presents the hotel name with a star rating and provides two buttons: Guest Login and Staff Login. This screen serves as the entry point for both user roles.

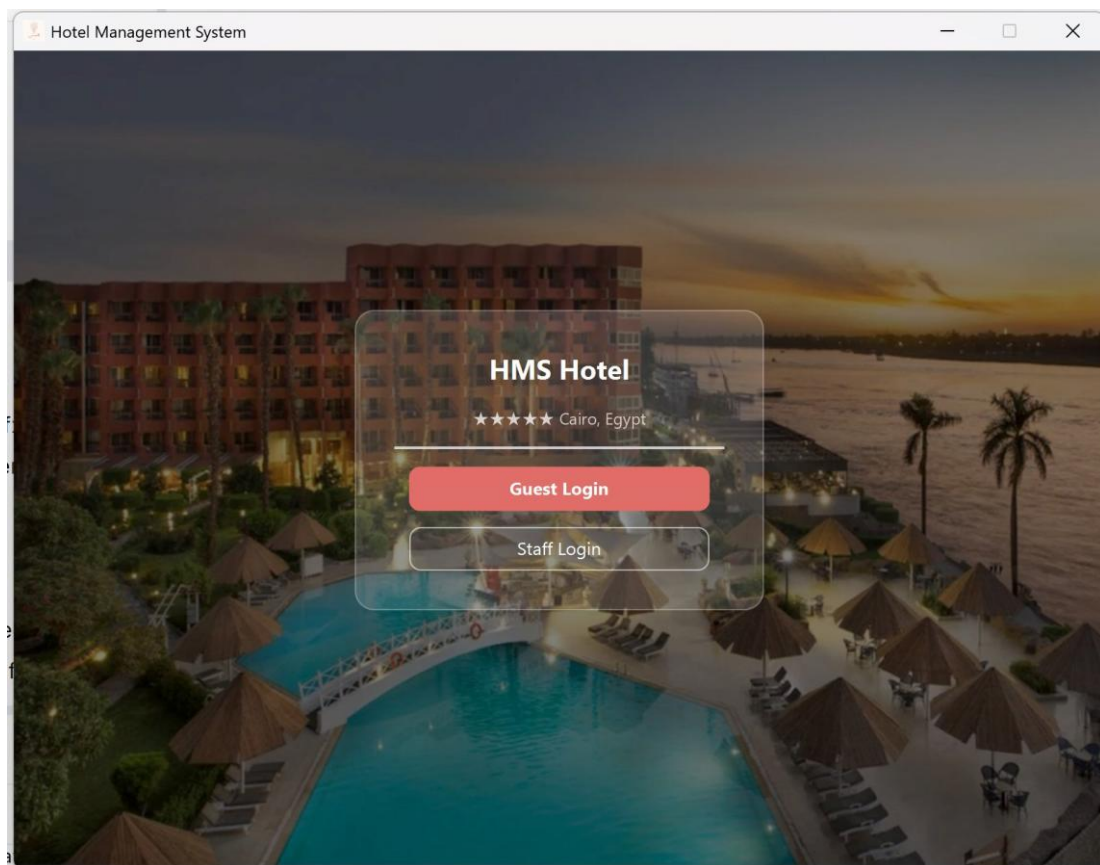


Figure 1: Welcome Screen – HMS Hotel landing page

3.3.2 Guest Login Screen

After selecting Guest Login, users are taken to the Hotel Login screen. It provides Username and Password fields, a Show Password checkbox for toggling password visibility, a Login button, a Sign Up button for new guests, and a Back button to return to the welcome screen.

Hotel Management System

Hotel Login

Username

Password

☐ Show Password

Login

Sign Up

Back

Figure 2: Guest Login Screen

3.3.3 Guest Registration Screen

New guests can register by clicking Sign Up. The Create Account form collects: Username, Password (min 6 characters), Confirm Password, Date of Birth (via date picker), Gender (dropdown: MALE/FEMALE), Address, and Initial Balance. Real-time validation provides immediate feedback – for example, a mismatch between password and confirm password shows the error message 'Passwords do not match.' in red below the form before the user even clicks Register. Additional validation prevents short passwords and empty fields.

Hotel – Register

×

Create Account

Username

mariam

Password

1234

Confirm Password

12345

☒ Show Password

Date of Birth

5/9/2025

Gender

FEMALE

Address

cccccccccccc

Initial Balance (\$)

1234444

Passwords do not match.

Register

← Back to Login

Figure 3: Registration Screen – Password mismatch validation

Hotel – Register

×

Create Account

Username

mariam

Password

12345

Confirm Password

12345

☒ Show Password

Date of Birth

5/9/2025

Gender

FEMALE

Address

cccccccccccc

Initial Balance (\$)

1234444

Registration failed: Password must be at least 6 characters

Register

← Back to Login

Figure 4: Registration Screen – Password length validation

3.3.4 Guest Dashboard

After a successful login, guests are directed to the Guest Dashboard. The dashboard features a warm red header bar displaying the hotel brand name, the guest's username, their account balance, and a Logout button. The left panel shows the guest's profile (avatar, name, gender, date of birth, address, gender, account balance) and a Make Reservation / Manage Reservations button. The right panel shows reservation statistics (Total, Pending, Confirmed, Cancelled) and a table of Active Reservations.

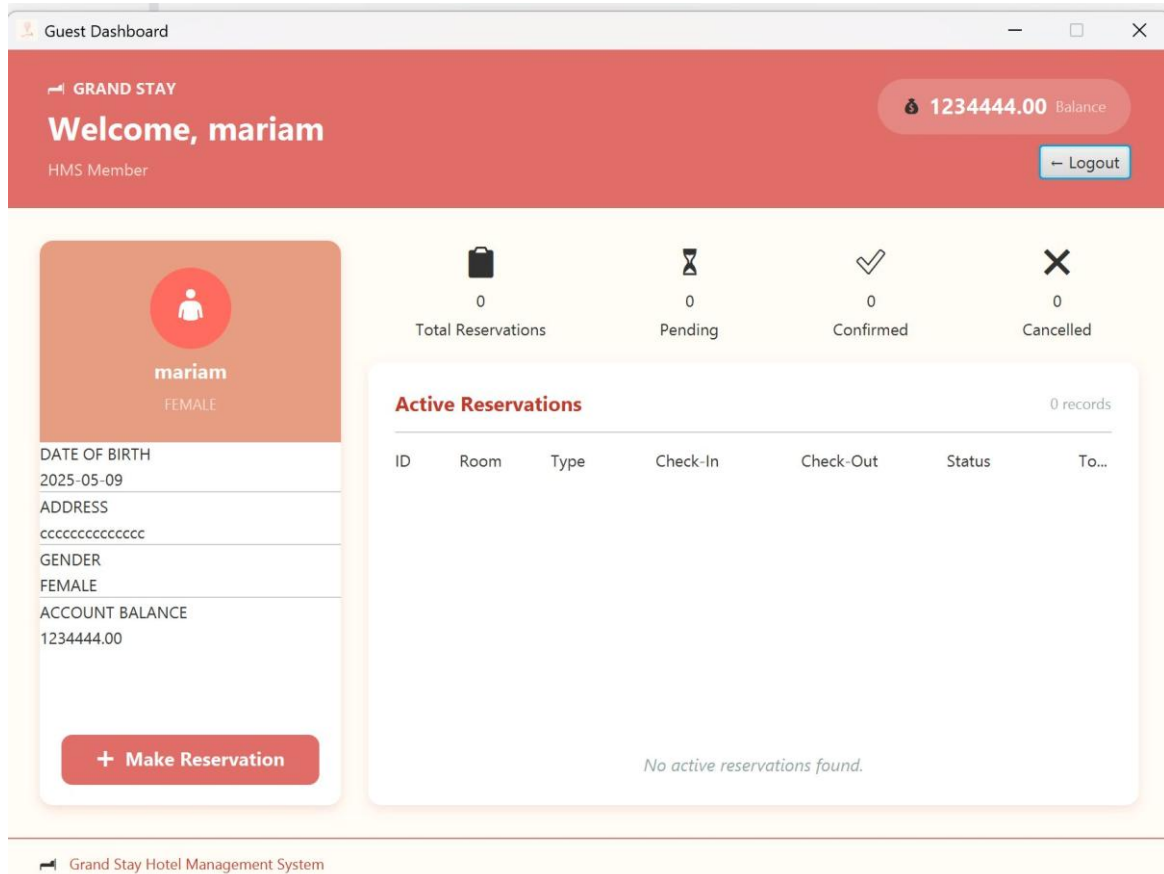


Figure 5: Guest Dashboard – Empty state (newly registered guest)

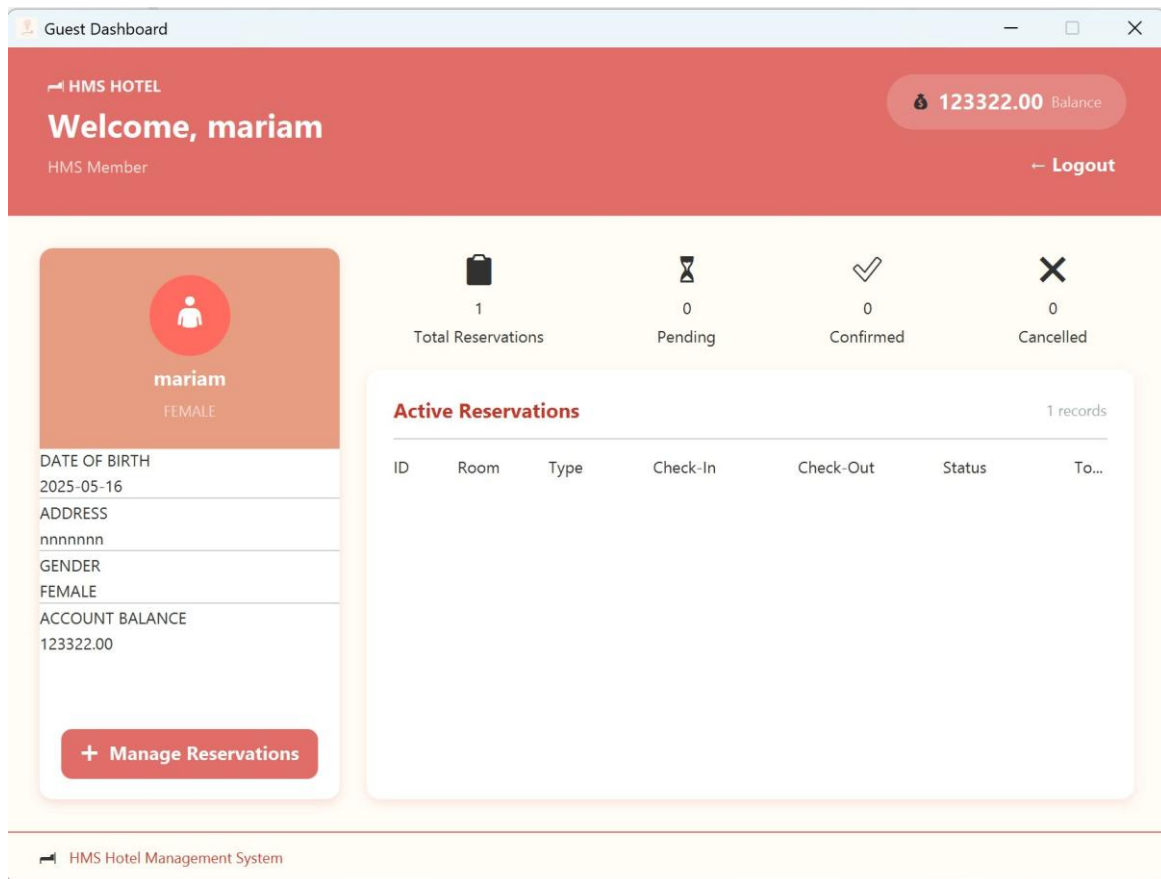


Figure 6: Guest Dashboard – Guest with 1 reservation

3.3.5 Make Reservation Screen

Guests can make a reservation by selecting their preferred Room Type (dropdown), preferred View (dropdown), Check-in Date, and Check-out Date using date pickers. Clicking Search Available Rooms filters and displays matching available rooms.

The screenshot shows a web browser window titled "Make Reservation". The main heading is "Make A Reservation" with the subtitle "Fill in your preferences and find available rooms". The form contains the following fields:

- Room Type:** A dropdown menu with the text "Select room type".
- View:** A dropdown menu with the text "Select preferred view".
- Check-in Date:** A date picker with the text "Select date".
- Check-out Date:** A date picker with the text "Select date".

At the bottom of the form are two buttons: "Back" and "Search Available Rooms". The "Search Available Rooms" button is highlighted in red. In the bottom right corner, it says "Logged in as: mariam".

Figure 7: Make Reservation – Filter form

3.3.6 Available Rooms Screen

After searching, the Available Rooms screen displays all matching rooms in a list (showing room number, type, view, floor, and price per night). On the right, guests can optionally select amenities (WiFi – \$50, TV – \$30, Mini Bar – \$100, Air Conditioning – \$80, Room Service – \$120) using Ctrl+Click for multiple selections. The header shows the applied filters and number of rooms found. Clicking Confirm Reservation books the selected room with chosen amenities.

Make Reservation

Available Rooms

Double | POOL_VIEW | 2026-05-09 → 2026-05-29 | 5 room(s) found

Select a Room

Room 104 Double POOL_VIEW Floor 4 \$950.0/night
Room 127 Double POOL_VIEW Floor 3 \$950.0/night
Room 143 Double POOL_VIEW Floor 1 \$950.0/night
Room 147 Double POOL_VIEW Floor 3 \$950.0/night
Room 217 Double POOL_VIEW Floor 2 \$950.0/night

Select Amenities (optional)

Hold Ctrl to select multiple

WiFi — \$50.0
TV — \$30.0
Mini Bar — \$100.0
Air Conditioning — \$80.0
Room Service — \$120.0

← Back

Confirm Reservation

Logged in as: —

Figure 8: Available Rooms – Room and amenity selection

3.3.7 Manage Reservations Screen

The Manage Reservations screen shows the guest's complete reservation history in a table with columns: ID, Room Number, Room Type, Check-In Date, Check-Out Date, and Reservation Status. Action buttons at the bottom allow the guest to Make Reservation, Cancel Reservation, or Check Out.

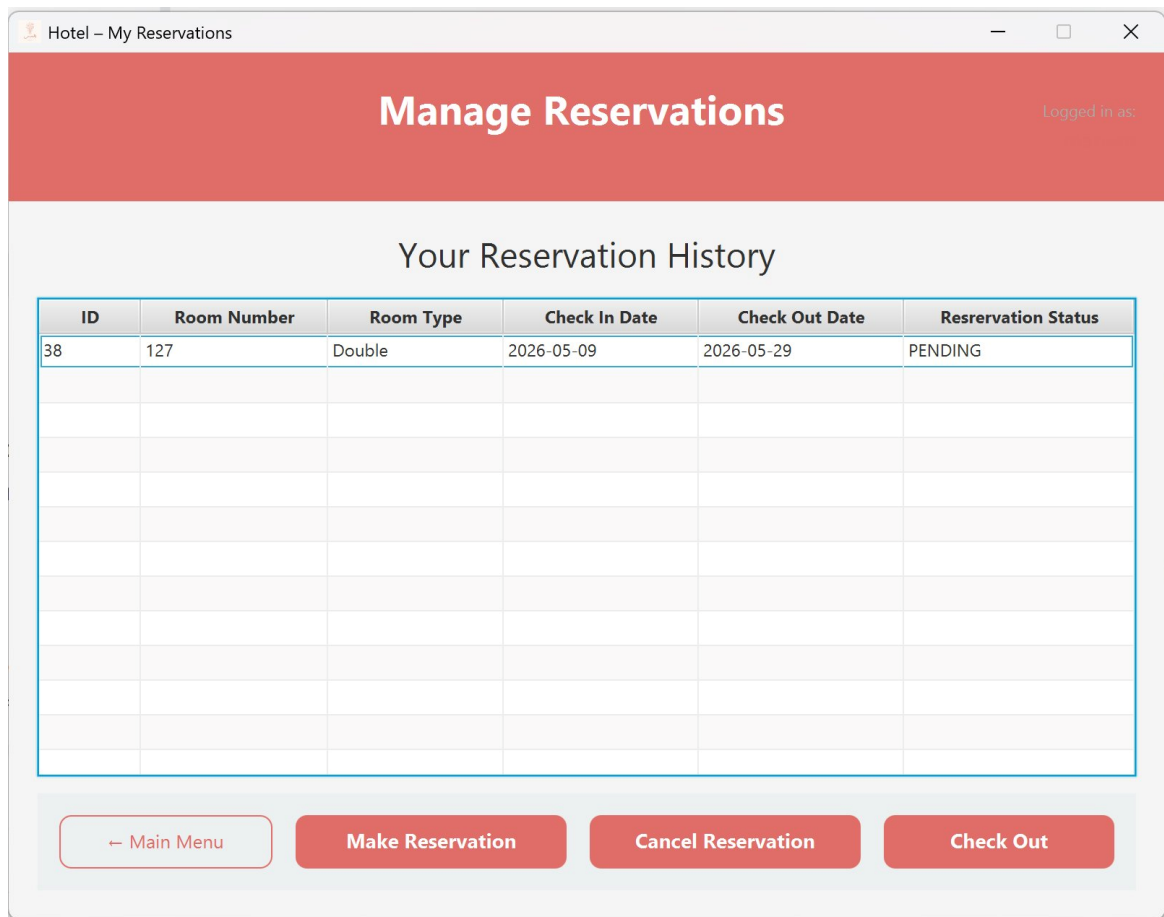


Figure 9: Manage Reservations – Reservation history with one active booking

3.3.8 Cancel Reservation Screen

Guests can cancel a reservation by selecting it from the active reservations list (highlighted in blue when selected) and clicking Confirm Cancellation. Upon success, the reservation disappears from the table and a green success message 'Reservation cancelled successfully.' is displayed.

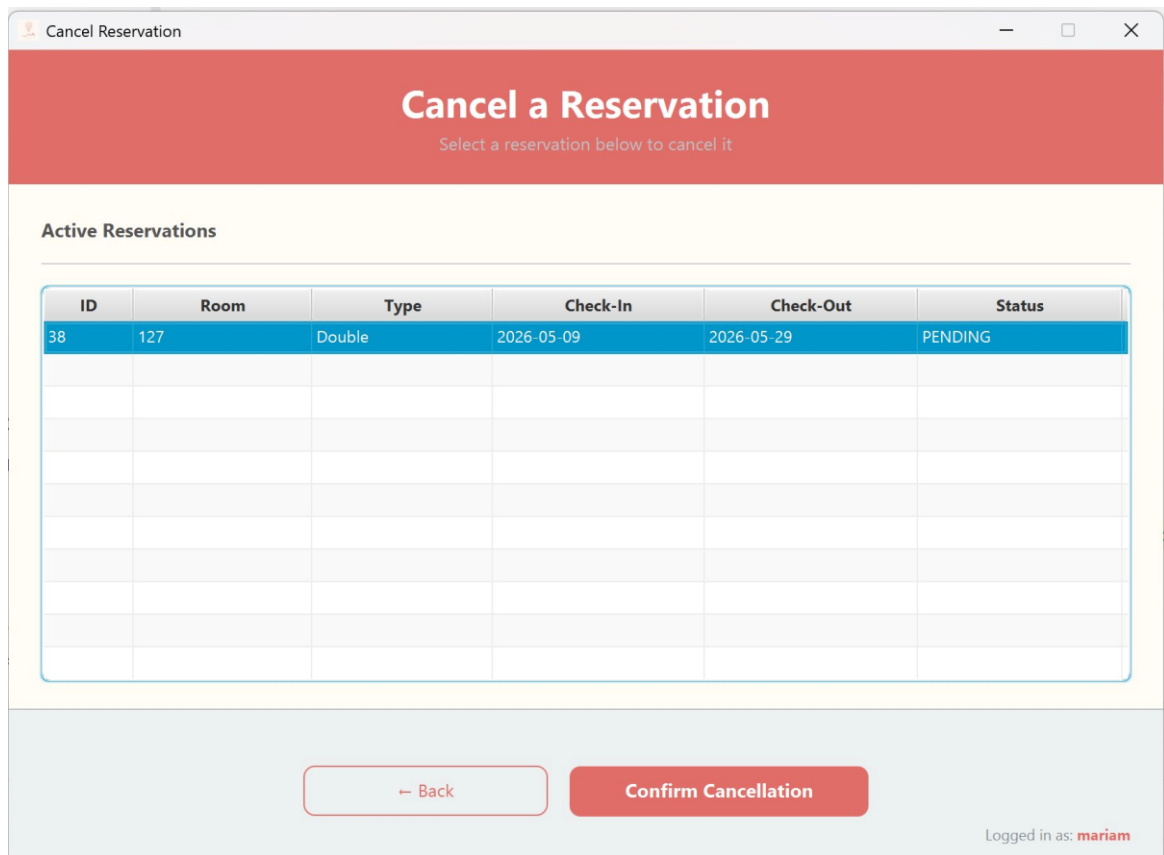


Figure 10: Cancel Reservation – Reservation selected and ready to cancel

3.3.9 Checkout Screen

The Checkout screen is accessible from the Manage Reservations screen. It shows currently checked-in reservations in a table on the left, and a Checkout Details panel on the right where the guest selects the room number and payment method (CASH, CREDIT_CARD, or ONLINE). The Payment Summary displays Room Cost, Amenities cost, and the Total Due. The guest's current balance is shown in green. Clicking Confirm Checkout processes the payment.

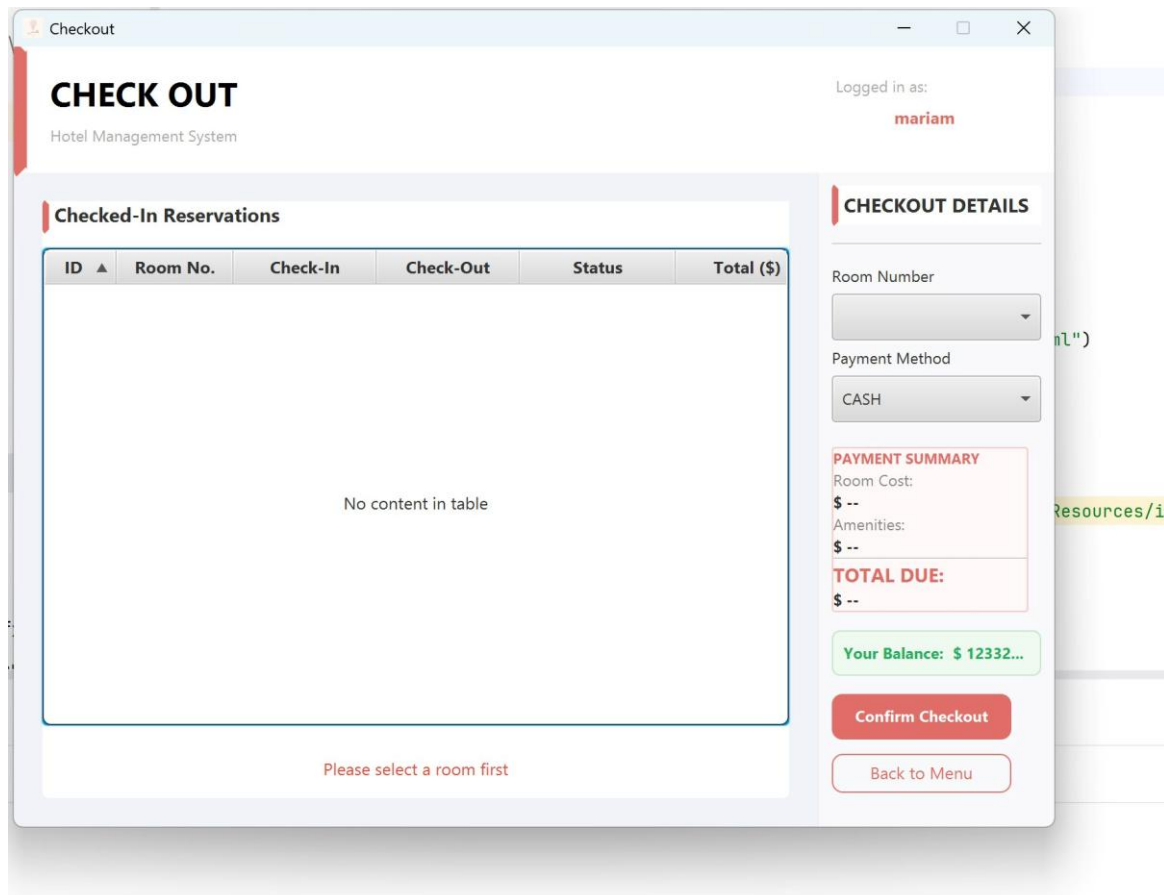


Figure 11: Checkout Screen – Payment summary and confirmation

3.3.10 Staff Login Screen

Staff members (Admin and Receptionist) access the system through a separate Staff Login screen reached via the Staff Login button on the welcome screen. The screen has a clean card-based design with Username and Password fields. Upon successful authentication, the system identifies the staff role and routes to either the Receptionist Dashboard or the Admin Dashboard accordingly.

HMS Hotel- Welcome



Staff Login

Username

Password

Login

Back

Figure 12: Staff Login Screen

3.3.11 Receptionist Dashboard

The Receptionist Dashboard provides a comprehensive sidebar with action buttons: All Reservations, Pending Reservations, Confirm Reservation, Cancel Reservation, Check In Guest, Process Checkout, Complete Reservation, View All Guests, and View All Rooms. The main content area displays a full reservation table showing all reservations with their ID, Guest, Room, Check-In, Check-Out, Status, and Total amount. Above the table, input fields allow the receptionist to enter a Reservation ID, Guest Username, Room Number, and Payment Method to perform the selected action.

Receptionist Dashboard

Receptionist Dashboard

HMS Hotel Management System

Logged in as:

Logout

ACTIONS

All Reservations

Pending Reservations

Confirm Reservation

Cancel Reservation

Check In Guest

Process Checkout

Complete Reservati...

View All Guests

View All Rooms

Reservation ID

Guest Username

Room Number

Payment Method

e.g. 5

e.g. ali_hassan_3

e.g. 105

Select method

Reservations

ID	Guest	Room	Check-In	Check-Out	Status	Tot
1	hassan_shawky_9	193	2026-05-14	2026-05-19	CANCELLED	3100
2	hala_rashad_25	158	2026-05-17	2026-05-21	CANCELLED	8480
3	ahmed_farouk_28	189	2026-05-14	2026-05-17	PENDING	1800
4	reem_fathy_14	173	2026-05-12	2026-05-15	CANCELLED	6300
5	hassan_shawky_9	229	2026-05-19	2026-05-21	PENDING	1800
6	farah_youssef_19	157	2026-05-11	2026-05-12	CONFIRMED	2200
7	karim_hassan_11	169	2026-05-12	2026-05-16	CANCELLED	3680
8	salma_farouk_10	140	2026-05-19	2026-05-24	CONFIRMED	1050
9	nour_adel_6	227	2026-05-11	2026-05-12	CONFIRMED	7000
10	reem_fathy_14	121	2026-05-12	2026-05-17	PENDING	6100
11	ibrahim_ali_17	135	2026-05-19	2026-05-24	CONFIRMED	3100
12	ibrahim_ali_17	200	2026-05-10	2026-05-12	CANCELLED	2400
13	salma_farouk_10	154	2026-05-11	2026-05-15	PENDING	3600
14	ali_mostafa_3	220	2026-05-14	2026-05-15	PENDING	2120

Figure 13: Receptionist Dashboard – All reservations view

Receptionist Dashboard

Receptionist Dashboard

HMS Hotel Management System

Logged in as:

Logout

ACTIONS

All Reservations

Pending Reservations

Confirm Reservation

Cancel Reservation

Check In Guest

Process Checkout

Complete Reservati...

View All Guests

View All Rooms

Reservation ID

Guest Username

Room Number

Payment Method

9

mariam

100

CASH

✓ Reservation #9 confirmed.

Reservations

ID	Guest	Room	Check-In	Check-Out	Status	Tot
1	hassan_shawky_9	193	2026-05-14	2026-05-19	CANCELLED	3100
2	hala_rashad_25	158	2026-05-17	2026-05-21	CANCELLED	8480
3	ahmed_farouk_28	189	2026-05-14	2026-05-17	PENDING	1800
4	reem_fathy_14	173	2026-05-12	2026-05-15	CANCELLED	6300
5	hassan_shawky_9	229	2026-05-19	2026-05-21	PENDING	1800
6	farah_youssef_19	157	2026-05-11	2026-05-12	CONFIRMED	2200
7	karim_hassan_11	169	2026-05-12	2026-05-16	CANCELLED	3680
8	salma_farouk_10	140	2026-05-19	2026-05-24	CONFIRMED	1050
9	nour_adel_6	227	2026-05-11	2026-05-12	CONFIRMED	7000
10	reem_fathy_14	121	2026-05-12	2026-05-17	PENDING	6100
11	ibrahim_ali_17	135	2026-05-19	2026-05-24	CONFIRMED	3100
12	ibrahim_ali_17	200	2026-05-10	2026-05-12	CANCELLED	2400
13	salma_farouk_10	154	2026-05-11	2026-05-15	PENDING	3600
14	ali_mostafa_3	220	2026-05-14	2026-05-15	PENDING	2120

Figure 14: Receptionist Dashboard – Successful reservation confirmation (green feedback)

Receptionist Dashboard

Receptionist Dashboard

HMS Hotel Management System

Logged in as:

Logout

ACTIONS

All Reservations

Pending Reservations

Confirm Reservation

Cancel Reservation

Check In Guest

Process Checkout

Complete Reservati...

View All Guests

View All Rooms

Reservation ID

Guest Username

Room Number

Payment Method

9

mariam

100

CASH

✖ Cannot complete! Reservation #9 is not CHECKED_OUT. Current status: CONFIRMED

Reservations

ID	Guest	Room	Check-In	Check-Out	Status	Tot
1	hassan_shawky_9	193	2026-05-14	2026-05-19	CANCELLED	3100
2	hala_rashad_25	158	2026-05-17	2026-05-21	CANCELLED	8480
3	ahmed_farouk_28	189	2026-05-14	2026-05-17	PENDING	1800
4	reem_fathy_14	173	2026-05-12	2026-05-15	CANCELLED	6300
5	hassan_shawky_9	229	2026-05-19	2026-05-21	PENDING	1800
6	farah_youssef_19	157	2026-05-11	2026-05-12	CONFIRMED	2200
7	karim_hassan_11	169	2026-05-12	2026-05-16	CANCELLED	3680
8	salma_farouk_10	140	2026-05-19	2026-05-24	CONFIRMED	1050
9	nour_adel_6	227	2026-05-11	2026-05-12	CONFIRMED	700.0
10	reem_fathy_14	121	2026-05-12	2026-05-17	PENDING	6100
11	ibrahim_ali_17	135	2026-05-19	2026-05-24	CONFIRMED	3100
12	ibrahim_ali_17	200	2026-05-10	2026-05-12	CANCELLED	2400
13	salma_farouk_10	154	2026-05-11	2026-05-15	PENDING	3600
14	ali_mostafa_3	220	2026-05-14	2026-05-15	PENDING	2120

Figure 15: Receptionist Dashboard – Error feedback for invalid operation (red message)

3.3.12 View All Guests Screen

The receptionist can open a separate View All Guests window displaying a full table of all 30 registered guests, showing Username, Address, Gender, Date of Birth, Balance, and number of Reservations per guest. This provides the receptionist with a complete overview of the guest database.

All Guests						
All Guests - Total: 30						
Username	Address	Gender	Date of Birth	Balance	Reservations	
hana_mahmoud_0	Cairo	FEMALE	1975-12-11	3419.0	0	
mohamed_hamdy_1	Cairo	MALE	1981-09-19	3158.0	2	
sara_aziz_2	Cairo	FEMALE	1983-12-20	2059.0	0	
ali_mostafa_3	Cairo	MALE	1972-10-16	4451.0	1	
fatma_youssef_4	Cairo	FEMALE	1995-06-20	3360.0	0	
youssef_fathy_5	Cairo	MALE	1991-09-19	3089.0	1	
nour_adel_6	Cairo	FEMALE	1994-10-03	5009.0	2	
omar_farouk_7	Cairo	MALE	1974-02-14	1562.0	2	
layla_nasser_8	Cairo	FEMALE	1971-08-23	2283.0	0	
hassan_shawky_9	Cairo	MALE	1973-03-06	2160.0	3	
salma_farouk_10	Cairo	FEMALE	1977-04-19	4543.0	2	
karim_hassan_11	Cairo	MALE	1981-05-16	4948.0	3	
rana_farouk_12	Cairo	FEMALE	1992-10-26	4916.0	1	
tarek_adel_13	Cairo	MALE	1999-05-09	1995.0	1	
reem_fathy_14	Cairo	FEMALE	1981-06-03	5840.0	3	
khaled_zaki_15	Cairo	MALE	1991-11-23	3433.0	0	
lina_youssef_16	Cairo	FEMALE	1981-12-26	1410.0	1	
ibrahim_ali_17	Cairo	MALE	1995-02-08	1544.0	2	
jana_mostafa_18	Cairo	FEMALE	1975-11-19	1231.0	1	
farah_youssef_19	Cairo	MALE	1974-01-03	4796.0	2	
mostafa_mahmoud_20	Cairo	MALE	1994-11-26	1736.0	3	
yasmin_rashad_21	Cairo	FEMALE	1978-01-15	4860.0	1	
amr_nasser_22	Cairo	MALE	1977-02-22	4647.0	1	
dina_aziz_23	Cairo	FEMALE	1978-07-08	5294.0	0	
mahmoud_youssef_24	Cairo	MALE	1980-02-17	3055.0	1	
hala_rashad_25	Cairo	FEMALE	1978-12-26	2536.0	2	

Figure 16: All Guests – Complete guest list (30 guests)

3.3.13 Admin Dashboard

The Admin Dashboard shares the same layout structure as the Receptionist Dashboard but provides management-focused actions. The sidebar contains four management options: Add Room, Add Room Type, Add Amenity, and Add Receptionist. Each option loads the corresponding form in the main content area.

The screenshot shows the 'Admin Dashboard' interface for a 'Hotel Management System — Admin Panel'. The user is logged in as 'admin1'. On the left, a 'MANAGEMENT' sidebar contains buttons for 'Add Room', 'Add Room Type', 'Add Amenity', and 'Add Receptionist'. The main content area features the 'Add New Room' form. This form includes input fields for 'Room Number' (with a placeholder 'e.g. 201') and 'Floor (0 - 5)' (with a placeholder 'e.g. 2'). It also has dropdown menus for 'Room Type' (labeled 'Select type') and 'View' (labeled 'Select view'). A red 'Add Room' button is positioned at the bottom right of the form.

Figure 17: Admin Dashboard – Add New Room form

This screenshot shows the same 'Admin Dashboard' interface after a successful room addition. A green confirmation message, 'Room 70 added successfully.', is displayed above the 'Add New Room' form. The form itself remains identical to the previous state, with the same input fields and dropdown menus. The 'Add Room' button is still present at the bottom right of the form.

Figure 18: Admin Dashboard – Room added successfully (green confirmation)

Admin Dashboard

Admin Dashboard

Hotel Management System — Admin Panel

Logged in as: **admin1** Logout

MANAGEMENT

Add Room

Add Room Type

Add Amenity

Add Receptionist

Room type 'penthouse' added successfully.

Add New Room Type

Category Name

Capacity

e.g. Penthouse

e.g. 5

Base Price (\$)

e.g. 1500

Add Room Type

Figure 19: Admin Dashboard – Add New Room Type form

Admin Dashboard

Admin Dashboard

Hotel Management System — Admin Panel

Logged in as: **admin1** Logout

MANAGEMENT

Add Room

Add Room Type

Add Amenity

Add Receptionist

Amenity 'wifi' added successfully.

Add New Amenity

Amenity Name

Price (\$)

e.g. Jacuzzi

e.g. 200

Add Amenity

Figure 20: Admin Dashboard – Add New Amenity form

The screenshot displays the Admin Dashboard interface. At the top, a red header bar contains the 'Admin Dashboard' title and 'Hotel Management System — Admin Panel' on the left, and 'Logged in as: admin1' with a 'Logout' link on the right. A left sidebar under the 'MANAGEMENT' section lists four buttons: 'Add Room', 'Add Room Type', 'Add Amenity', and 'Add Receptionist'. The main content area features a green success message: 'Receptionist 'recertionist2' added successfully.' Below this is the 'Add New Receptionist' form, which includes four input fields: 'Username' (with placeholder 'e.g. receptionist3'), 'Password' (with placeholder 'Min. 6 characters'), 'Date of Birth' (with a date picker and placeholder 'Select date'), and 'Working Hours / Day' (with placeholder 'e.g. 8'). A red 'Add Receptionist' button is positioned at the bottom right of the form.

Figure 21: Admin Dashboard – Add New Receptionist form

4. UI Requirements Fulfillment

4.1 JavaFX Layouts

All screens use proper JavaFX layout containers as required. The application makes use of `BorderPane` for the main dashboards, `VBox` and `HBox` for vertical/horizontal groupings of controls, `GridPane` for form layouts (login, register, checkout details), `StackPane` layers for the welcome screen overlay effect, and `TableView` for all data tables.

4.2 FXML Files

All screens in the application are defined using FXML layout files, satisfying the requirement of at least 3 FXML files. Each screen has its own dedicated FXML file with a corresponding controller class, following the MVC separation pattern. The FXML files include but are not limited to:

- `Welcome.fxml` – Landing screen with background image and login buttons
- `GuestLogin.fxml` – Guest authentication form
- `Register.fxml` – New guest registration form
- `GuestDashboard.fxml` – Guest home screen with profile and reservation summary
- `MakeReservation.fxml` – Room search preferences form
- `AvailableRooms.fxml` – Room and amenity selection
- `ManageReservations.fxml` – Reservation history table
- `CancelReservation.fxml` – Cancellation screen
- `Checkout.fxml` – Payment and checkout screen
- `StaffLogin.fxml` – Staff authentication form

- ReceptionistDashboard.fxml – Receptionist operations panel
- AdminDashboard.fxml – Admin management panel

4.3 CSS Styling

A single external CSS stylesheet (styles.css) is applied to all scenes in the application. The stylesheet establishes a consistent warm color theme using the primary accent color #E06D67 (coral red) and background #FFF5F5 (warm off-white). Input fields have rounded corners (border-radius: 8px), buttons provide hover feedback by darkening the background color, and error messages are styled in red using the .status-label class. This creates a cohesive professional look across all screens.

4.4 User Feedback and Alerts

All user actions produce appropriate feedback messages displayed inline within the screens. This was implemented without using separate dialog pop-ups in most cases, keeping the UX clean:

- Green success messages (e.g., 'Room 70 added successfully.', 'Reservation #9 confirmed.', 'Reservation cancelled successfully.') appear above or below the relevant form area.
- Red error messages (e.g., 'Passwords do not match.', 'Registration failed: Password must be at least 6 characters', 'Cannot complete! Reservation #9 is not CHECKED_OUT.') provide specific actionable feedback.
- Status labels update in real-time as the user interacts with forms.

4.5 Scene Navigation

Navigation between screens is handled through JavaFX Scene switching. Each controller holds a reference to the Stage and calls `stage.setScene(new Scene(...))` when navigating to a new screen. Back buttons are consistently present to allow users to return to the previous screen. The Receptionist and Admin dashboards use in-place content switching (loading different form panels into the main content area) rather than creating new windows, reducing visual disruption.

5. Design Decisions

5.1 Single CSS File Approach

A single shared stylesheet was chosen over per-screen CSS files to ensure visual consistency and minimize maintenance. All style classes are reusable across screens, which means any future design change only needs to be made in one place.

5.2 Inline Feedback vs. Alert Dialogs

Rather than using JavaFX Alert dialog pop-ups for every validation message, the team chose to display feedback as styled labels within the screen itself. This provides a less disruptive user experience and keeps the user focused on the form they are filling in.

5.3 Role-Based Navigation

The system detects the logged-in user's role (ADMIN vs. RECEPTIONIST) after staff login and automatically routes them to the appropriate dashboard. This role-based routing prevents a receptionist from accidentally accessing admin-only features such as adding room types or managing amenities.

5.4 In-Memory Data Binding

All GUI components (TableView, ComboBox, ListView) are data-bound directly to the in-memory HotelDatabase static lists from Milestone 1. This means that when an admin adds a new room, it immediately appears in the guest's room search results without any additional synchronization code.

5.5 Show Password Toggle

Both the Guest Login and Register screens include a 'Show Password' checkbox. This was implemented by toggling between a PasswordField (masked) and a TextField (visible) while keeping the entered text synchronized, improving usability especially on longer passwords.

6. Test Cases

The following test cases were designed and verified to validate the correct behavior of the system under both valid and invalid scenarios.

6.1 Authentication Test Cases

Test Case	Description	Input	Expected Output	Result
TC-A01	Guest login with valid credentials	Username: mariam, Password: 123456	Navigates to Guest Dashboard	Pass
TC-A02	Guest login with wrong password	Username: mariam, Password: wrongpass	Error message shown	Pass
TC-A03	Register with mismatched passwords	Password: 1234, Confirm: 12345	Shows 'Passwords do not match.'	Pass
TC-A04	Register with short password	Password: 12345 (5 chars)	Shows 'Password must be at least 6 characters'	Pass
TC-A05	Register with all valid inputs	All fields filled correctly	Account created, redirected to login	Pass
TC-A06	Staff login as receptionist	Valid receptionist credentials	Navigates to Receptionist Dashboard	Pass
TC-A07	Staff login as admin	Valid admin credentials (admin1)	Navigates to Admin Dashboard	Pass

6.2 Reservation Test Cases

Test Case	Description	Input	Expected Output	Result
TC-R01	Search for available rooms with valid filters	Type: Double, View: POOL_VIEW, Dates: valid range	5 rooms listed with prices	Pass
TC-R02	Make a reservation for an available room	Select Room 127, no amenities, confirm	Reservation created with PENDING status	Pass
TC-R03	Cancel an existing PENDING reservation	Select reservation ID 38, click Confirm Cancellation	Reservation removed, success message shown	Pass
TC-R04	Attempt to complete a non-CHECKED_OUT reservation	Reservation #9 is CONFIRMED	Error: 'Reservation #9 is not CHECKED_OUT'	Pass
TC-R05	Receptionist confirms a PENDING reservation	Reservation ID: 9, Guest: mariam, Room: 100	Status changes to CONFIRMED, green message shown	Pass
TC-R06	View reservation history	Guest with existing booking	Table shows reservation with all details	Pass

6.3 Admin Management Test Cases

Test Case	Description	Input	Expected Output	Result
TC-M01	Admin adds a new room	Room 70, Floor 2, valid type and view	Success: 'Room 70 added successfully.'	Pass
TC-M02	Admin adds a new room type	Category: penthouse, Capacity: 2, Base: 2000	Success: 'Room type penthouse added successfully.'	Pass
TC-M03	Admin adds a new amenity	Name: wifi, Price: 50	Success: 'Amenity wifi added successfully.'	Pass
TC-M04	Admin adds a new receptionist	All fields filled with valid data	Success message shown	Pass
TC-M05	View all guests as receptionist	Click View All Guests	Window shows 30 guests with all details	Pass

7. Limitations

Although the system successfully covers all required Milestone 2 features, several limitations exist in the current version:

- **In-Memory Storage Only:** All data is stored in static ArrayLists in the HotelDatabase class and is lost when the application is closed. A real persistent database (e.g., MySQL) would be needed for production use.
- **No Multi-threading:** The current implementation runs all operations on the JavaFX Application Thread. For large datasets or network-bound operations, background threads would be needed to prevent UI freezing.
- **No Networking:** The bonus client-server chat feature was not implemented in this milestone.
- **No Image Management:** Admin cannot upload room photos through the GUI.

- Limited Admin Panel: The admin can only add records; update and delete operations for rooms, room types, and amenities are not yet available through the GUI.

8. Conclusion

Milestone 2 successfully extended the Hotel Management System backend from Milestone 1 into a fully functional, visually polished JavaFX desktop application. All mandatory requirements were met: a complete JavaFX GUI with FXML layout files for all screens, a consistent CSS stylesheet, role-based navigation for guests and staff, and proper user feedback throughout the application.

Through this milestone, the team gained practical experience in JavaFX application development, FXML-based layout design, CSS theming, and MVC-style controller architecture. The project demonstrates how OOP principles from Milestone 1 translate directly into a clean, maintainable GUI application.

The final system provides a solid foundation that can be extended in future work with database integration, multi-threading, and networking features.